Out-of-Pocket Healthcare Expenditure Analysis: Identifying Household and District-Level Drivers in India

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Abstract

This study analyzes out-of-pocket (OOP) healthcare expenditure in India using a large-scale citizen survey ($n = 49,819$) to identify drivers of financial hardship. Utilizing Lasso regularization ($\lambda = 0.001335$), we identify facility type as the primary predictor of expenditure intensity; private facility use adds $\beta = 6.38$ to log-OOP, an effect size significantly larger than all other demographic variables. Paradoxically, Logistic GLM analysis reveals that insured households face higher catastrophic odds ($OR = 1.53, p \approx 0$), likely driven by adverse selection rather than facility choice. Furthermore, Quadratic Logistic GLM confirms a non-linear relationship between household consumption and risk, identifying a high-risk plateau across the bottom half of the consumption distribution where meaningful financial protection only materializes at the top quartile. These findings suggest that OOP intensity is overwhelmingly determined by facility choice, indicating that current insurance instruments provide insufficient depth against private sector pricing and chronic illness shocks.

> "Of all the forms of inequality, injustice in health is the most shocking and the most inhuman because it often results in physical death."
> - **Martin Luther King Jr.**

1 INTRODUCTION & MOTIVATION

Out-of-Pocket Expenditure (OOPE) on Health refers to direct payments made by individuals to healthcare providers at the time of service, which are not covered by insurance, government schemes, or third-party payers. OOP spending remains a critical driver of financial hardship and a primary barrier to healthcare equity.

1.1 The Gap Between National Aggregate Trends and Household Reality

While the Government of India reports a significant structural shift in health financing, with public expenditure increasing from 29.0% to 48.0% and aggregate out-of-pocket (OOP) spending declining from 62.6% to 39.4% [1]- it remains unclear if these macro-level trends translate into meaningful financial protection at the household level. Despite a national budgetary allocation of approximately **9,500 crore INR** for FY 2026-27, our analysis reveals a negligible correlation between insurance coverage and absolute OOP levels ($r = -0.005$). This disconnect is further emphasized by our finding that insured households paradoxically face higher catastrophic expenditure odds ($OR = 1.53, p \approx 0$). This "Insurance Paradox" suggests that while public spending has lowered the national average, the current reimbursement ceilings fail to shield sicker, private-facility-reliant populations from financial ruin. Consequently, while the government expands the breadth of enrollment, true financial protection only materializes at the top income quartile, leaving the bottom 50% of the consumption distribution exposed to nearly identical levels of catastrophic risk.

1.2 Research Questions

Following this motivation, this study addresses five core research questions:

- **RQ1:** Which household characteristics are the most significant predictors of OOP expenditure?
- **RQ2:** Does insurance coverage reduce OOP expenditure across different population subgroups?
- **RQ3:** How does the OOP gap between public and private facilities vary across care types?
- **RQ4:** Does a non-linear threshold exist between consumption and catastrophic risk?
- **RQ5:** Do district-level infrastructure factors explain residual variation in spending?

Apart from the Research Questions mentioned above, we also analyze State-Level OOP Spending Variation, Caste & Gender Disparities, Insurance-Facility Selection to further evaluate the "Insurance Paradox" as mentioned above.

1.3 Scope of Study

This study analyzes a large-scale observational dataset derived from a citizen health survey ($n = 49,819$). The analysis focuses on both household-level predictors - such as chronic illness (NCD) status, insurance enrollment, and facility choice - and district-level infrastructure characteristics, including PHC density and specialist availability. By integrating these multi-level factors, the study aims to move beyond aggregate metrics to identify the specific inflection points where healthcare costs become financially ruinous for Indian households.

2 DATA DESCRIPTION

2.1 Source and Collection

The dataset for this project has been sourced from the Citizen Survey Dataset, India published on Harvard Dataverse [2]. The Citizens Survey was conducted by the Lancet Commission on a Citizen-Centred Health System for India. The data were collected from November 2022 to April 2023 by interviewing respondents in person in 50,000 randomly selected households across 125 districts in 29 Indian states and Union Territories. The survey comprised 141 questions covering healthcare utilization, experiences, costs, satisfaction, delivery preferences, insurance coverage, willingness to pay, health information behaviors, technology use, aspirational health norms, and electoral attitudes towards health.

2.1.1 Raw Data Summary

The raw dataset (`.dta` format, converted to `.csv`) and its variable codebook were downloaded together, as column names are encoded (e.g., `a2a`, `b2b`). Table 2.1.1 presents the pre-processing snapshot.

Parameter	Value
Observations (Rows)	50,217
Features (Columns)	232
Total data-points	11,650,344
Duplicate rows	0
Constant columns	4
High-null columns (>80%)	30
Numerical features	180
Categorical features	52
High-cardinality columns (>50 levels)	22

2.2 Pre-processing Pipeline

The raw dataset required extensive cleaning before analysis. A four-stage pipeline was applied systematically. The four-stage pipeline has been described below:

1. **Variable Selection and Type Coercion**:

   The raw dataset contains 232 columns, many of which are administrative identifiers (e.g., household codes, interviewer IDs), free-text fields, and variables with no analytical relevance to OOP expenditure. These were dropped to reduce noise and dimensionality. Column names in the raw file are encoded short codes derived from the survey instrument (e.g., a2g for food expenditure, b4 for outpatient visit indicator, c5 for inpatient admission). These were mapped to interpretable names using the provided variable code-book. Numeric expenditure columns were explicitly coerced to float type to prevent silent misclassifications of values as strings. The insurance column required standardisation: survey waves encoded non-coverage inconsistently as "0", blank, or "No", all of which were unified to a single "No" category. Four constant columns (zero variance across all respondents) were dropped as they carry no information. Thirty columns with more than 80% missing values were also removed. Respondents who did not provide consent (consent_code $\neq 1$) were filtered out, reducing the sample from 50,217 to 50,000.

2. **Missing Data Resolution**:

   The dataset exhibited two distinct forms of missingness and measurement artifacts:

   - **Structural Missingness:** Inpatient and outpatient out-of-pocket (OOP) expenditures were missing for respondents with no recorded healthcare utilization. These were imputed as \$0\$, as the absence of an admission or visit implies zero cost by construction.
   - **Non-structural Missingness:** Food expenditure, education expenditure, and household size contained six missing values. These were resolved via median imputation to ensure robustness against right-skewed distributions and extreme outliers.
   - **Data Cleaning:** Two negative inpatient OOP values, interpreted as reimbursement artifacts (where insurance settlements exceeded gross expenditure), were clipped to zero to maintain economic consistency.

3. **Feature Engineering**:

   Three central variables were constructed: Total OOP expenditure (sum of inpatient and outpatient costs) and total household consumption (sum of food and education expenditure), the latter serving as a proxy for economic status (Section 2.4). A catastrophic expenditure indicator was defined as 1 if total OOP exceeded 10% of consumption.

   To accommodate zero-expenditure observations, both variables were transformed using $\log(1 + x)$. Facility types were classified into Private, Public, or Other via keyword mapping. Categorical predictors-including occupation, caste, age, and consumption quartiles-were converted to dummy variables.

   Notably, education levels were consolidated from five categories into three (Low, Medium, High) to resolve multicollinearity identified via Variance Inflation Factor (VIF) analysis (Section 3.1)

4. **Finalization and Standardization**:

   After feature engineering, the dataset was checked for residual duplicates (none found) and constant columns (six additional constant dummies removed).

   The final analytical dataset contains 50,000 rows and 269 columns, with memory usage reduced from 544.99 MB to approximately 35 MB. Numerical features used in LASSO were standardized to zero mean and unit variance prior to model fitting, as the L1 penalty is scale-sensitive.

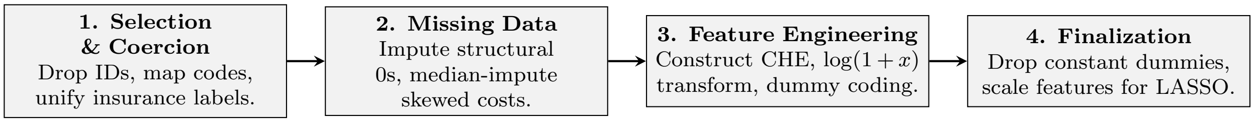


Figure 1: Four-stage Pre-processing Pipeline

Table 1: Summary of Primary Variables used in Modelling

Variable Name	Type	Role	Description
`total_oop`	Continuous	Primary response	Total OOP in rupees (IP + OP)
`log_oop`	Continuous	Transformed response	log(1 + total_oop)
`catastrophic`	Binary	Binary response	1 if OOP > 10% of consumption
`total_consumption`	Continuous	Key predictor	Food + education expenditure (Rs.)
`log_consumption`	Continuous	Transformed predictor	log(1 + total_consumption)
`any_insurance`	Binary	Predictor / control	1 if any insurance coverage
`facility_op_type`	Categorical	Predictor	Private / Public / Other
`residence_Urban`	Binary	Predictor	1 if urban household
`uhc_index`	Continuous	District predictor	Composite infrastructure score
`ncd_op / ncd_ip`	Binary	Predictor	1 if NCD illness (chronic)
`had_op_visit`	Binary	Derived	1 if outpatient visit occurred
`had_ip_admit`	Binary	Derived	1 if inpatient admission occurred
`district_name`	Categorical	Fixed effect	District identifier (29 states)

2.3 Key Variables

The analytical framework centers on three variations of the healthcare expenditure response variable, alongside socioeconomic predictors and district-level controls. The primary response, total_oop, is modeled both in its raw form and as a log-transformed continuous variable log_oop to stabilize variance. Additionally, the binary *catastrophic* indicator identifies households where health spending exceeds 10% of total consumption.

Key predictors include total_consumption (a proxy for economic status) and facility_op_type. We control for any_insurance and residence_Urban (urbanicity) to isolate the impact of healthcare access on spending. Finally, district_name is incorporated as a fixed-effect variable to account for regional heterogeneity across 29 states, as discussed in the context of Research Question 5.

2.4 Constructed Variables and Proxy Logic

Catastrophic Expenditure Indicator. Following World Health Organization (WHO) standards, a binary catastrophic health expenditure (CHE) indicator is defined using a 10% threshold of total household consumption. Formally:

$$CHE_i = \begin{cases} 1 & \text{if total\_oop}_i > 0.10 \times \text{total\_consumption}_i \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

This threshold identifies spending levels that likely crowd out essential non-medical needs. In rare instances where consumption is zero, any positive OOP is classified as catastrophic ($CHE = 1$).

Consumption Proxy. Total household consumption is proxied by the sum of food and education expenditures. Consumption is preferred over income in the Indian context due to the high prevalence of informal labor and self-employment, which render income reporting volatile and prone to measurement error. While this proxy may underestimate total economic capacity by excluding housing and utilities, it provides a stable and reliable metric for benchmarking living standards.

Negative OOP and Distributional Adjustments. Two negative inpatient OOP values (minimum: -Rs. 26,500) were identified as reimbursement artifacts - cases where insurance settlements exceeded gross expenditures. These were clipped to zero to maintain economic interpretability within an OOP framework.

Due to the extreme right-skewness of expenditure data (median $= 530$; mean $= 4,049$), variables were transformed using the $\log(1+x)$ convention. This stabilizes residual variance for OLS estimation and accommodates the high mass of zero-OOP observations (over 50% for inpatient data) while allowing for multiplicative coefficient interpretation ($e^{\beta}-1$).

Note: Coefficients in the following log-linear models are interpreted multiplicatively. For a coefficient β, the associated percentage change in the response variable for a unit increase in the predictor is calculated as $(e^{\beta}-1) \times 100\%$.

3 METHODOLOGY

3.1 RQ1: Log-Linear OLS with LASSO

Out-of-pocket (OOP) healthcare expenditure exhibits extreme right-skewness; over 50% of respondents report zero inpatient costs, while the upper tail reaches millions. This distribution violates OLS assumptions regarding residual normality and homoscedasticity. We apply a $\log(1 + x)$ transformation to stabilize variance and approximate symmetry (Figure 2).
The log-linear model is specified as:

$$\log(1 + OOP_i) = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} + \epsilon_i \tag{2}$$

To manage 33 candidate predictors and prevent overfitting, we employ **Lasso** regularization. Lasso adds an L_1 penalty to the OLS objective:

$$\min_{\beta} \left[\sum_i (y_i - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right] \tag{3}$$

The optimal penalty ($\lambda = 0.001335$) is selected via 10-fold cross-validation. Post-selection, the model is re-fitted using OLS with HC3 robust standard errors. Multicollinearity was resolved by consolidating education levels until all Variance Inflation Factors (VIF) fell below 10.

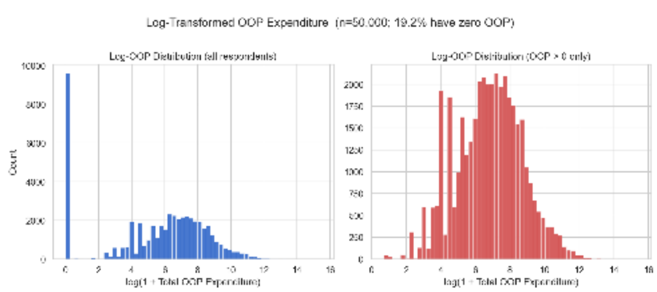


Figure 2: $log(1 + x)$ Transformation on OOPE

3.2 RQ2: Logistic GLM for Catastrophic Expenditure

For the binary catastrophic expenditure indicator, we utilize the Generalized Linear Model (GLM) framework with a logit link to bound predicted probabilities p_i within $[0, 1]$:

$$\ln\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} \tag{4}$$

Parameters are estimated via Maximum Likelihood Estimation (MLE). Coefficients are interpreted as **Odds Ratios (OR)**. Heterogeneous protection is tested through nested models with interaction terms (e.g., Ins $\times$ Quartile), evaluated via a Likelihood-Ratio Test (LRT). We also apply Welch's t-tests to compare mean log-OOP between insurance groups without assuming equal variances.

3.3 RQ3: OLS with Facility $\times$ Episode Interaction

The data is restructured into a long-format dataset where each row represents a discrete episode. We utilize log-linear OLS rather than a Poisson GLM, as OOP is a continuous monetary value. The interaction model is:

$$\begin{aligned} \log(1 + OOP_{ep}) &= \beta_0 + \beta_1 \text{Priv} + \beta_2 \text{Inpat} \\ &+ \beta_3 (\text{Priv} \times \text{Inpat}) + \gamma' \mathbf{z} + \epsilon \end{aligned} \tag{5}$$

The term β_3 captures the "private premium" specific to inpatient care, testing if private-sector pricing becomes more aggressive for complex episodes.

3.4 RQ4: Threshold Detection via Quadratic GLM

To detect non-linear "threshold effects" between consumption and risk, we implement a quadratic logistic GLM. This allows the relationship to bend, capturing plateaus in risk at low-income levels:

$$\begin{aligned} \text{logit}(p_i) &= \beta_0 + \beta_1 \log C_i \\ &+ \beta_2 (\log C_i)^2 + \gamma' \mathbf{z} \end{aligned} \tag{6}$$

A significant quadratic term (β_2), confirmed via LRT ($\chi^2(1) = 263.0, p < 0.001$), provides statistical evidence for the threshold effect.

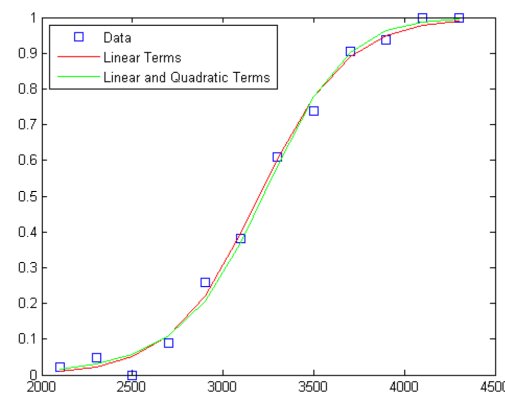


Figure 3: Data Fitting with GLMs

3.5 RQ5: District Infrastructure and Partial F-Test

To account for supply-side factors, we incorporate the district-level Universal Health Coverage (UHC) index. We use a nested MLR framework to test if district infrastructure adds explanatory power beyond household-level controls:

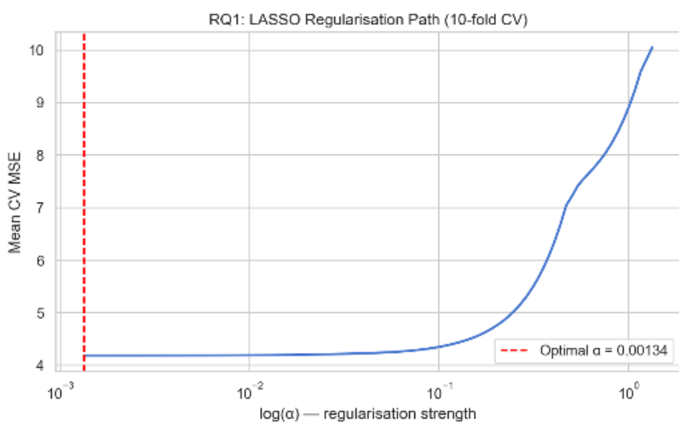
$$F = \frac{(RSS_{rest} - RSS_{full})/q}{RSS_{full}/(n - k_{full} - 1)} \tag{7}$$

where $q = 1$ (the UHC parameter). A significant F-statistic implies that regional resources independently reduce household financial burden.

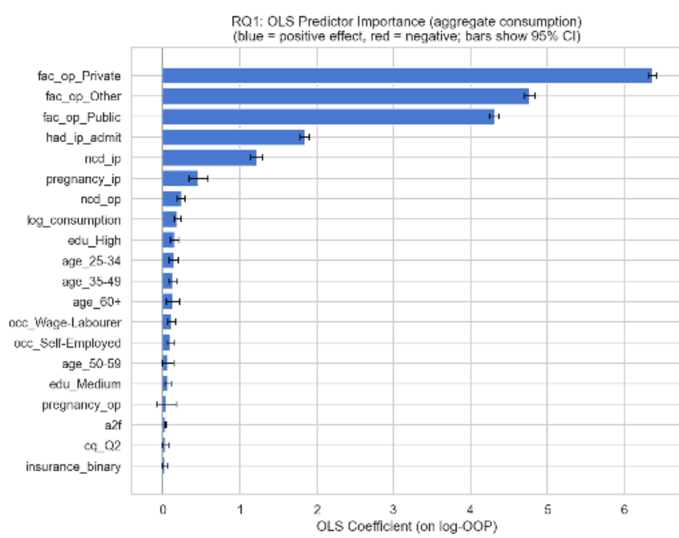
4 RESULTS AND ANALYSIS

4.1 Predictors of OOP (RQ1)

The Lasso procedure selected **32 of 33** candidate features at the **optimal penalty** ($\lambda = 0.001335$), achieving a test-set R^2 **of 0.590**.



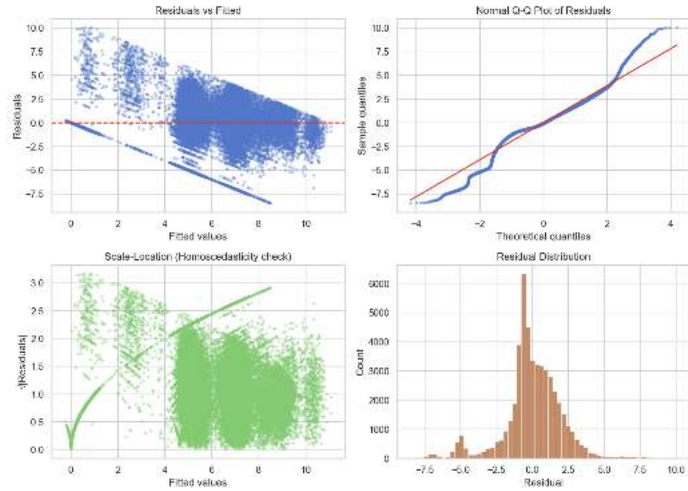
Subsequent OLS re-estimation on these features yielded an adjusted $R^2 = 0.586$ ($F = 3,453.88, p < 0.001$). This high explanatory power suggests that the selected predictors capture nearly 60% of the variation in log-OOP expenditure.



Facility type emerges as the overwhelming determinant of expenditure. **Private facility use** contributes $\beta = 6.38$ to log-OOP (95% CI: $[6.32, 6.44]$), an effect size that dwarfs all other predictors by a factor of 30. Other major drivers include **inpatient admission** ($\beta = 1.94$) and NCD-related episodes ($\beta = 1.29$). While socioeconomic factors like **log-consumption** ($\beta = 0.20$) and advanced age are statistically significant, their influence is secondary to the choice of healthcare provider. **Sensitivity analyses** excluding top-tier outliers and utilizing per-capita metrics confirmed the robustness of these rankings.

4.1.1 Model Diagnostics

To evaluate the validity of the OLS assumptions, we performed a four-part residual diagnostic analysis.



- **Residuals vs. Fitted:** This plot evaluates linearity; the distinct diagonal boundary represents the "floor" created by structural zeros in the expenditure data. The concentration of residuals suggests that the log-linear specification successfully captures the functional relationship across various scales.
- **Normal Q-Q Plot:** The Q-Q plot compares error distribution against a theoretical normal line to validate p-value reliability. While the S-curve indicates heavy tails from extreme outliers and structural zeros, the central alignment suggests the log-transform has sufficiently normalized the data for OLS inference.
- **Scale-Location Plot:** This panel assesses homoscedasticity by plotting standardized residuals against fitted values. The non-random clustering confirms that error variance is not constant, a trait in healthcare financial data.
- **Residual Distribution:** The histogram provides a direct visualization of error symmetry, showing a dominant peak at zero with a roughly bell-shaped spread. This confirms that the $\log(1 + x)$ transformation successfully mitigated the extreme right-skew of the raw OOPE data.

Note: The optimal $\lambda = 0.001335$ is small because the large sample ($n = 50,000$) provides sufficient data to estimate 33 predictors without heavy regularization - the CV loss curve is flat near zero, and the bias-variance tradeoff favors minimal shrinkage. As $\lambda \to 0$, the LASSO solution converges to OLS; the penalty here serves primarily for feature selection rather than coefficient shrinkage.

4.2 Insurance Effectiveness (RQ2)

The main-effects logistic model (Model A) achieves a McFadden's R^2 of 0.139, representing a strong fit for social science data. The main-effects logistic model (Model A) achieves a McFadden's R^2 of 0.139, representing a strong fit for social science data. The model further achieves an AUC of **0.7302**, indicating good discriminative ability - the model correctly ranks a randomly selected catastrophic household above a non-catastrophic one 73% of the time. The interaction model (Model B) significantly improves fit: LRT $\chi^2(4) = 13.94$, $p = 0.007$, confirming that insurance effectiveness is heterogeneous across income groups. Key coefficients from Model A are presented in Table 2.

Predictor	OR	95% CI	p
Insurance	1.525	[1.460, 1.593]	<0.001
Private Facility	7.733	[7.352, 8.135]	<0.001
Urban Residence	0.868	[0.830, 0.908]	<0.001
Log Consumption	0.991	[0.921, 1.066]	0.803
Male Gender	1.093	[1.051, 1.136]	<0.001
NCD Outpatient	2.122	[2.010, 2.239]	<0.001
Quartile 2 vs. Q1	0.920	—	0.024
Quartile 3 vs. Q1	0.760	—	<0.001
Quartile 4 vs. Q1	0.494	—	<0.001

Table 2: Logistic GLM coefficients - (Model A)

The Insurance Paradox. Insured households face 52.5% *higher* odds of catastrophic expenditure ($OR = 1.525$, $p < 0.001$), the opposite of what insurance is designed to achieve. We investigated the mechanism directly in AA4: cross-tabulation shows insured households are actually *less* likely to use private facilities (28.1% vs. 39.3%; $\chi^2(1) = 407.8$, $p < 0.001$). Facility selection is therefore not the driver. The most defensible explanation is **adverse selection**: sicker individuals disproportionately enrol in insurance and incur high OOP regardless of coverage, because existing schemes (e.g., Ayushman Bharat) carry low reimbursement ceilings, exclude outpatient costs, and suffer claims processing delays. Table 3 confirms that insurance provides no meaningful catastrophic protection at either facility type.

Group	Private	Public
Insured	81.9%	53.3%
Uninsured	79.6%	44.5%

Table 3: Catastrophic rate (%) by insurance status and facility type

Income Gradient. The quartile gradient is monotonic and strong: moving from Q1 to Q4 halves catastrophic odds ($OR = 0.494$, $p < 0.001$). Economic capacity provides more reliable financial protection than current insurance designs. Log-consumption itself is not significant ($p = 0.803$) as a continuous predictor once quartile dummies capture the non-linear structure.

Welch's t-test. Insured households have significantly higher mean log-OOP: difference = 0.316, 95% CI [0.254, 0.378], $t(24{,}660) = 9.99$, $p = 1.9 \times 10^{-23}$.

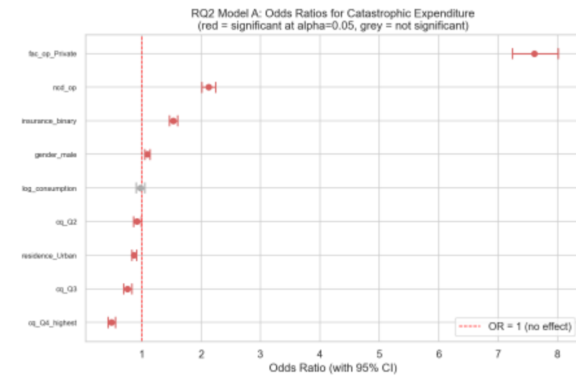


Figure 4: Odds ratios for catastrophic expenditure (Model A). Red points are statistically significant ($\alpha = 0.05$); grey points are not. The insurance OR of 1.53 lies to the right of 1, confirming the paradox.

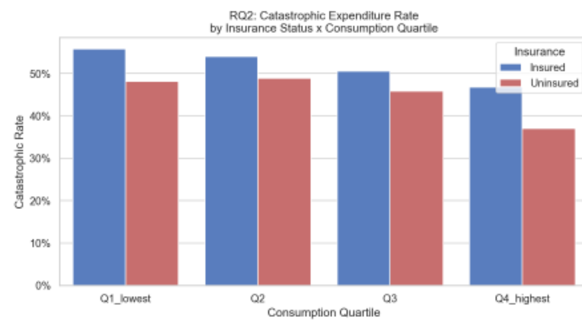


Figure 5: Catastrophic expenditure rate by insurance status across consumption quartiles. Insurance protection materialises only weakly, confirming that income quartile is the dominant buffer.

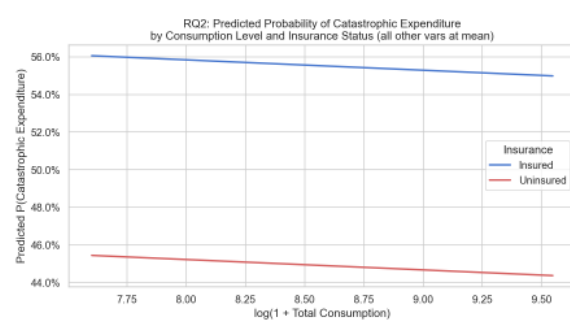


Figure 6: Predicted probability of catastrophic expenditure vs. log-consumption, holding all other variables at their means. The insured curve consistently exceeds the uninsured curve, consistent with adverse selection.

4.3 Facility Type vs. OOP (RQ3)

The long-format episode dataset contains 26,916 observations: 15,394 outpatient and 11,522 inpatient. Table 4 presents mean log-OOP by cell, showing a clear cost hierarchy in the episode type.

Facility	Episode	n	Mean	Median
Private	Inpatient	2,794	9.48	9.55
Private	Outpatient	4,882	7.58	7.71
Public	Inpatient	8,728	5.97	6.43
Public	Outpatient	10,512	5.24	5.77

Table 4: Mean and median log-OOP by facility type and episode type

The OLS interaction model ($R^2 = 0.277$, $F = 2{,}699.69$, $p < 0.001$) decomposes the cost gap as follows:

$$\log(1 + OOP) = 5.24 + 2.31\,\text{Private} + 0.72\,\text{Inpatient} + 1.17\,(\text{Private} \times \text{Inpatient}) \tag{8}$$

The three key estimates are:

- $\hat{\beta}_1 = 2.31$ - private premium for outpatient care over public
  ($p < 0.001$, 95% CI: [2.25, 2.37])
- $\hat{\beta}_2 = 0.72$ - inpatient premium at public facilities
  ($p < 0.001$, 95% CI: [0.65, 0.79])
- $\hat{\beta}_3 = 1.17$ - additional private premium for inpatient over outpatient
  ($p = 6.5 \times 10^{-132}$, 95% CI: [1.08, 1.27])

The interaction coefficient $\hat{\beta}_3$ is the central finding. The total private premium for outpatient episodes is 2.31 log-units; for inpatient episodes it is $2.31 + 1.17 = 3.48$ log-units. The private premium **nearly doubles** for inpatient care, consistent with unregulated pricing for surgical procedures, ICU stays, and implants where public alternatives are scarce. Welch's t-test on pooled episodes confirms the overall private-public gap: difference in mean log-OOP = 2.70, 95% CI [2.65, 2.75], $t(20{,}304) = 102.4$, $p < 0.001$.

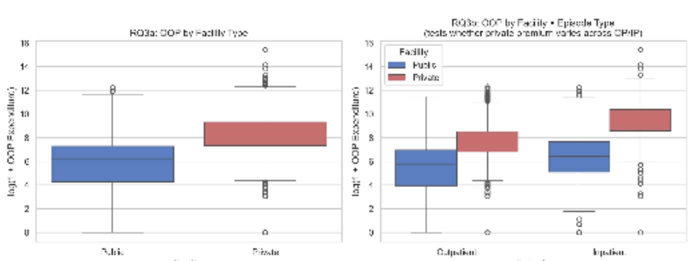


Figure 7: Log-OOP by facility type (left panel) and by facility × episode combination (right panel). The private-inpatient cell has the highest median and the narrowest spread, reflecting uniform high pricing by private hospitals.

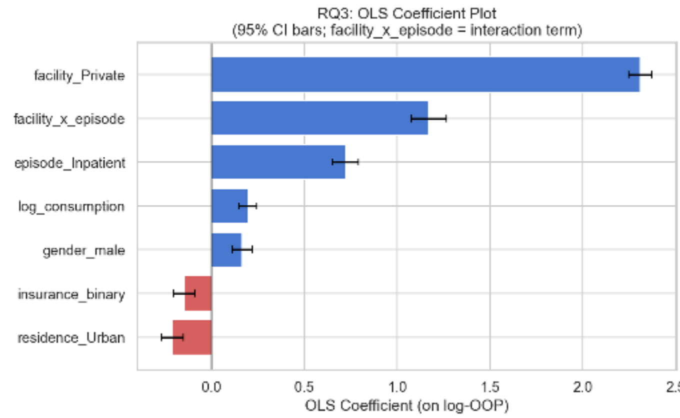


Figure 8: OLS coefficient estimates with 95% CI for RQ3. All three terms are highly significant. The interaction bar ($\hat{\beta}_3 = 1.17$) confirms that the private premium widens substantially for inpatient episodes.

4.4 Non-Linear Thresholds (RQ4)

The Likelihood Ratio Test (LRT) comparing the linear (Model A) and quadratic (Model B) logistic models yields $\Lambda = 263.04$, $\chi^2(1)$, $p = 3.7 \times 10^{-59}$. The quadratic term is overwhelmingly significant, providing strong statistical evidence for a non-linear threshold in the consumption - catastrophe relationship. **Model fit improves from McFadden's $R^2 = 0.025$ (linear) to 0.029 (quadratic).**

In Model B, the sign pattern ($\hat{\beta}_1 = +4.05$, $\hat{\beta}_2 = -0.25$) describes an inverted-U shape in log-odds: **catastrophic probability first rises with consumption, peaks, then falls. This reflects the structure of the catastrophic indicator itself - very low consumption households sometimes have low absolute OOP (because they forgo care entirely), while middle-consumption households have enough income to seek care but not enough to absorb the cost.** Financial protection only materialises consistently at the top consumption quartile.

Quartile	Q1	Q2	Q3	Q4
Catastrophic Rate	50.7%	50.8%	47.7%	40.1%
n	12,506	13,708	11,969	11,817

Table 5: Catastrophic expenditure rate by consumption quartile (Q1 = lowest, Q4 = highest)

The quartile table makes the shape concrete: Q1 and Q2 sit on an identical high-risk plateau (50.7% vs. 50.8%), while the decline only begins at Q3 (47.7%) and becomes substantial at Q4 (40.1%). The bottom half of the distribution is **equally exposed**. Policies targeting only the very poorest leave the second income quartile fully unprotected.

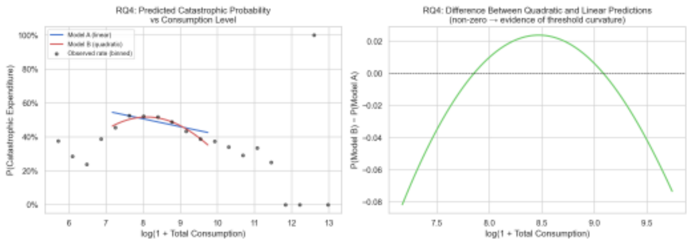


Figure 9: Predicted catastrophic probability vs. log-consumption. Left panel: linear (blue) vs. quadratic (red) model curves with observed binned rates (black dots). The quadratic curve better tracks the observed plateau-then-decline shape. Right panel: difference between models, confirming curvature at moderate consumption.

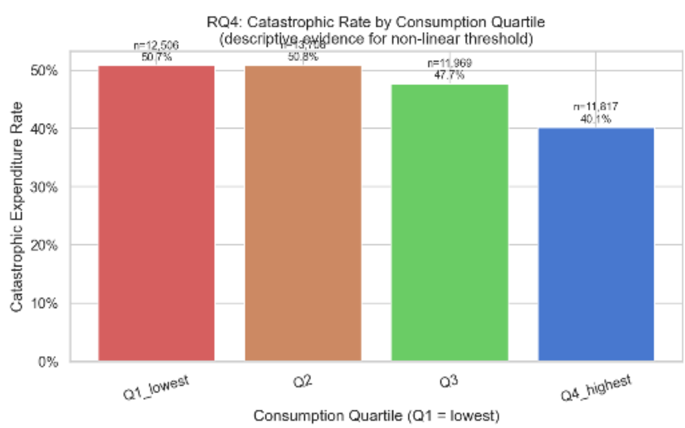


Figure 10: Catastrophic expenditure rate by consumption quartile. The flat Q1–Q2 plateau followed by a drop at Q3–Q4 visually confirms the threshold structure detected by the Likelihood Ratio Test.

4.5 District Infrastructure (RQ5)

Incorporating the district-level UHC index into a nested MLR framework significantly improves model fit beyond household controls alone. Table 6 summarises the two nested models estimated on the 46,200 respondents for whom the UHC index is available.

	Model A (HH controls)	**Model B** (+UHC index)
R^2	0.063	0.071
Adj. R^2	0.063	0.071
n	46,200	46,200
Partial F-test: $F(1,\,46192) = 425.77$, $p = 3.6 \times 10^{-94}$		
$\Delta R^2 = 0.009$; UHC $\hat{\beta} = -0.0433$, $p = 8.7 \times 10^{-98}$		

Table 6: Nested MLR comparison for RQ5. Model B adds the district UHC index.

The UHC index coefficient $\hat{\beta} = -0.0433$ implies that every one-unit improvement in a district's UHC index is associated with a 4.3% reduction in OOP. The index spans 36 units across our sample (30.7 to 66.7), meaning a household in the best-resourced district faces approximately $0.0433 \times 36 = 1.56$ fewer log-units of OOP than an identical household in the worst-resourced district, holding all personal characteristics constant. All VIFs in Model B remain below 2, confirming no multicollinearity from the district predictor. Tercile analysis shows that High-UHC districts have a mean catastrophic rate of 44.4% versus 49.7% in Low-UHC districts - a 5.3 percentage point reduction attributable to infrastructure quality alone.

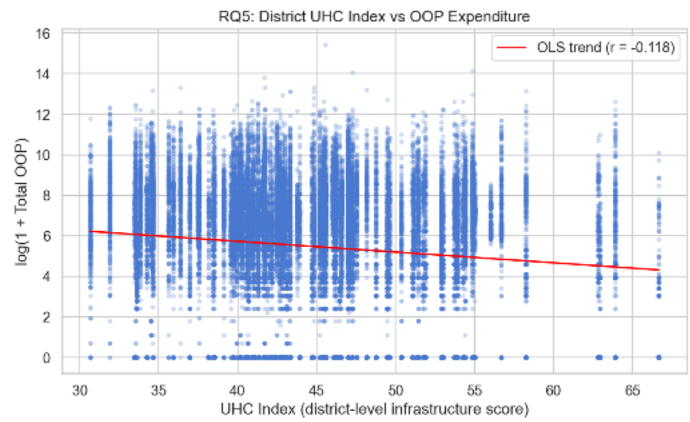


Figure 11: Scatter plot of district UHC index vs. log-OOP with OLS trend line. Better-resourced districts are significantly associated with lower household OOP independent of individual characteristics.

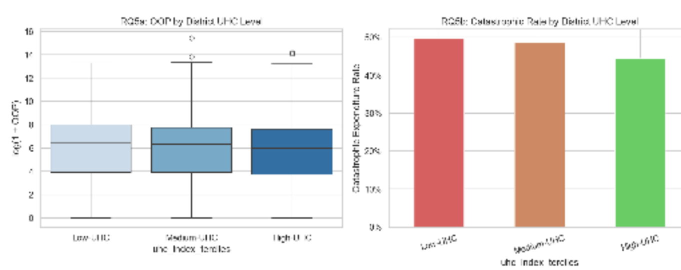


Figure 12: Log-OOP (left) and catastrophic rate (right) by UHC index tercile. High-UHC districts show both lower OOP levels and a 5.3 pp lower catastrophic rate than Low-UHC districts.

4.6 Additional Analyses (Beyond Requirement)

4.6.1 State-Level OOP Variation (AA1)

State membership alone explains 17.5% of OOP variation (ANOVA $R^2 = 0.175$, $F = 377.87$, $p < 0.001$). The spread of catastrophic rates across 29 states is 74 percentage points - from Sikkim (94.5%) to Arunachal Pradesh (20.6%). Table 7 shows the most and least affected states.

State	CHE Rate	Ins. Rate
Sikkim	94.5%	43.8%
Manipur	78.1%	8.3%
Meghalaya	71.0%	2.8%
Karnataka	69.8%	22.9%
Punjab	24.8%	30.7%
Nagaland	22.2%	4.1%
Arunachal Pradesh	20.6%	19.5%

Table 7: Top and bottom states by catastrophic expenditure (CHE) rate with corresponding insurance coverage rates

The insurance effect on catastrophic rates varies from −9.6 percentage points (insurance reduces risk, as expected) to +39.5 percentage points (insurance substantially increases risk). The mean insurance effect across states is +13.4 pp — in most states, insurance is associated with *higher* catastrophic risk, consistent with the adverse selection explanation from RQ2.

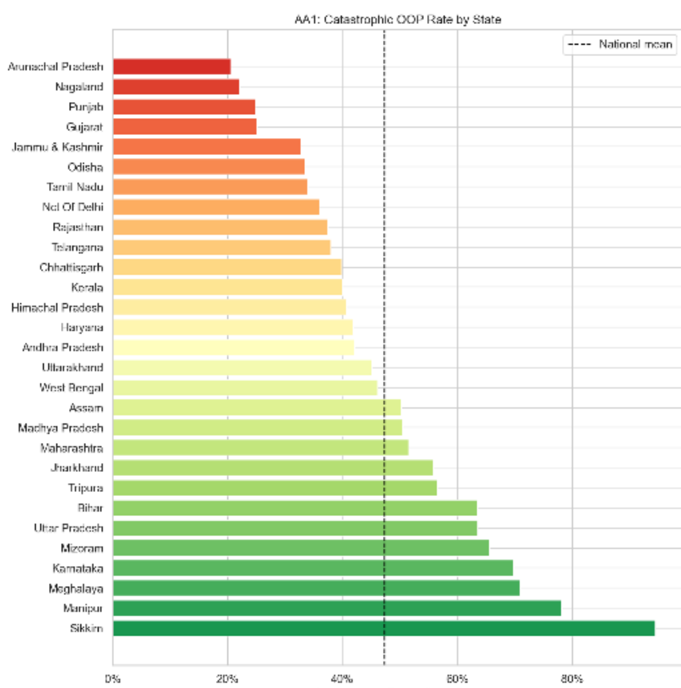


Figure 13: Catastrophic OOP rate across 29 Indian states, sorted highest to lowest. The dashed line shows the national mean (47.5%). The 74 percentage point range indicates that geography is as important as household characteristics in determining financial vulnerability.

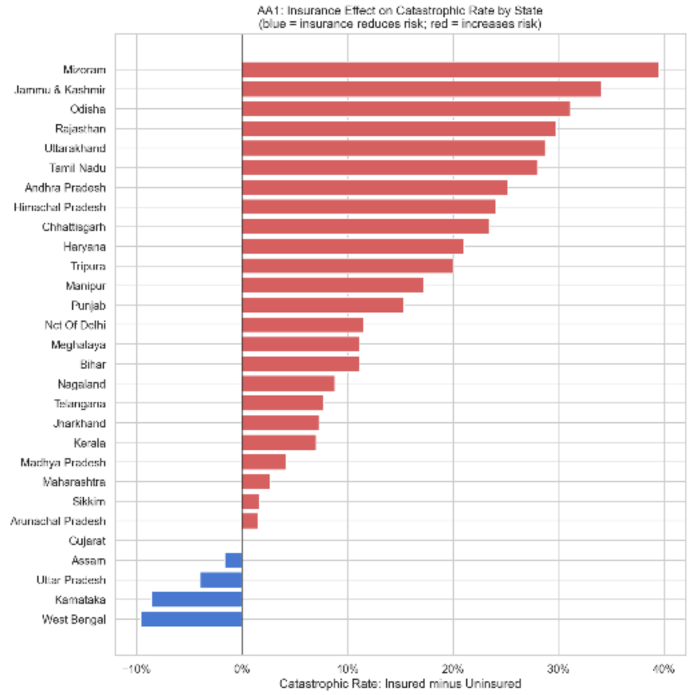


Figure 14: Insurance effect (insured minus uninsured catastrophic rate) by state. Blue bars (negative) indicate states where insurance reduces catastrophic risk; red bars (positive) indicate states where it increases risk. Most states show a positive (adverse) insurance effect.

4.6.2 Chronic Illness - NCD Double Burden (AA2 and AA5)

NCD Stands for Non-Communicable Diseases.

NCD outpatients incur substantially higher OOP than non-NCD patients (mean Rs. 8,132 vs. Rs. 4,673). Welch's t-test: difference in mean log-OOP = 0.680, 95% CI [0.592, 0.767], $t(5{,}588) = 15.26$, $p = 1.5 \times 10^{-51}$. Controlling for facility type, insurance, and SES in a logistic GLM, NCD status carries $OR = 2.11$ (95% CI [2.000, 2.228], $p < 0.001$). Chronic illness independently doubles catastrophic risk beyond care-seeking patterns.

A critical equity finding emerges from AA5: NCD patients are *less* likely to be insured (24.3% vs. 27.7%; $\chi^2(1) = 40.82$, $p < 0.001$). Table 8 shows the 2×2 catastrophic rate table — the most financially exposed subgroup is NCD + Uninsured (65.0%), which is 14.8 percentage points above Non-NCD + Insured (50.1%). Those who need financial protection most are the least likely to have it.

	Insured	Uninsured
NCD (Chronic)	67.3%	65.0%
Non-NCD (Acute)	50.1%	41.5%

Table 8: Catastrophic expenditure rate (%) by NCD status and insurance coverage. NCD patients face high catastrophic risk regardless of insurance.

Caste	n	% Any OOP	% OP Visit	% IP Admit
General	13,039	83.3%	35.4%	24.9%
OBCs	16,741	80.8%	31.4%	22.0%
SCs	10,328	85.2%	33.0%	24.6%
STs	9,231	72.5%	33.1%	21.9%

Table 9: Care-seeking rates by caste group. STs have the lowest rate of any healthcare spending (72.5%), supporting the constrained access hypothesis.

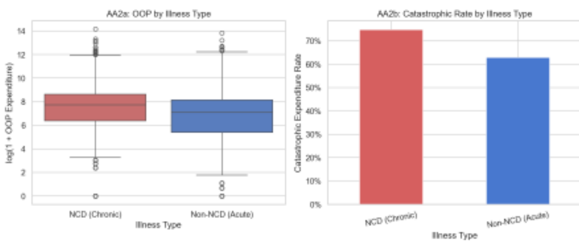


Figure 15: OOP distribution (left) and catastrophic rate (right) by illness type. NCD patients face substantially higher OOP and a 12.6 pp higher catastrophic rate than non-NCD patients.

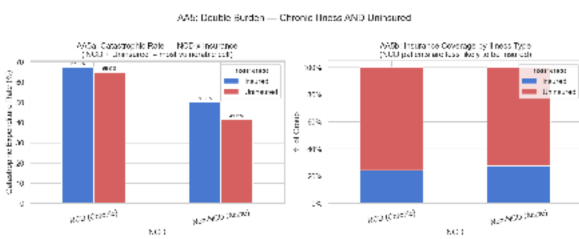


Figure 16: Left: catastrophic rate by NCD status × insurance (the double burden cell NCD+Uninsured at 65.0% is the highest). Right: insurance coverage rate by illness type - NCD patients are significantly less likely to be insured despite having greater need.

4.6.3 Caste Disparities and Constrained Access (AA3 and AA6)

SC/ST households have significantly lower log-OOP than General caste households: difference $= -0.540$, 95% CI $[-0.609, -0.471]$, $t(28{,}288) = -15.30$, $p = 1.2 \times 10^{-52}$. This appears to be a positive outcome but AA6 reveals it is not. When the comparison is restricted to households who *actually sought care* (OOP > 0), the gap narrows from -0.540 to -0.322 - a **40% reduction**. This narrowing is the key diagnostic: ST households have the lowest rate of any healthcare spending at all (72.5% having any OOP), versus 83.3% for General caste. SC/ST households are not spending less because they are healthier - they are seeking less care. Lower OOP partially reflects **foregone treatment** due to cost barriers, not a financial advantage. Table 9 summarises care-seeking rates by caste.

No significant gender difference in OOP is observed: $t(46{,}957) = -0.254$, $p = 0.799$. Healthcare spending is gender-neutral in this dataset.

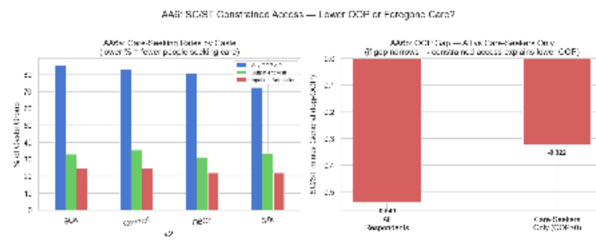


Figure 17: Left: care-seeking rates (any OOP, outpatient visit, inpatient admission) by caste group. Right: SC/ST vs. General log-OOP gap in the full sample vs. care-seekers only. The 40% narrowing confirms constrained access.

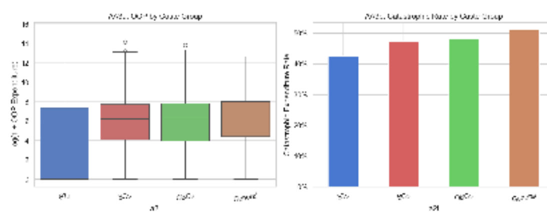


Figure 18: OOP expenditure (left) and catastrophic rate (right) by caste group. Despite lower absolute OOP, SC/ST households face catastrophic rates comparable to General caste once care-seeking differences are accounted for.

4.6.4 Deconstruction of Insurance Paradox with Numbers (AA4)

To explain why insured households face higher catastrophic odds ($OR = 1.53$), we analyzed facility selection. Paradoxically, insured households are **less likely** to use private facilities (28.1% vs. 39.3%; $p \approx 0$), yet those who do face the highest catastrophic risk at 81.9%. This confirms the paradox is driven by **adverse selection** - where sicker individuals enroll in insurance that lacks the depth to cover private sector pricing - rather than a simple preference for expensive care.

Insurance \ Facility	Private	Public
Insured	28.06	71.94
Uninsured	39.33	60.67

Table 10: Insurance Status by Facility Selection (Row %)

5 BENCHMARKING AGAINST STATE OF THE ART & PRIOR LITERATURE

Our findings are situated within five highly authoritative sources: the Government of India's National Health Accounts, two high-impact peer-reviewed studies in the WHO Bulletin and the International Journal for Equity in Health, a recent clinical validation study on Ayushman Bharat published in BMC Medicine, and the dataset's own Nature Scientific Data paper.

5.1 National Health Accounts 2021–22 (MoHFW)

The Ministry of Health and Family Welfare released the National Health Accounts (NHA) for 2021–22 in September 2024 [7]. The NHA reports that out-of-pocket expenditure has declined from 62.6% of total health expenditure (THE) in 2013–14 to 39.4% in 2021–22, and per capita OOP stands at Rs. 1,657 at current prices.

Convergence. Both the NHA and our study confirm that OOP remains a dominant health financing burden. The NHA reports approximately 90 million Indians breach the catastrophic threshold annually; our household-level data shows 47.5% of surveyed households do so.

Divergence. Our mean per-respondent OOP of Rs. 4,049 exceeds the NHA per capita figure of Rs. 1,657. Three structural differences explain this. First, the NHA divides total national OOP spending by the entire population, including non-users of healthcare; our survey captures only households across 125 healthcare-active districts. Second, our consumption proxy (food + education only) underestimates the true denominator, increasing the fraction classified as catastrophic. Third, our data (November 2022 to April 2023) postdates the NHA reference year.

What our study adds. The NHA provides macro accounting but cannot identify which household characteristics drive OOP. Our LASSO + OLS framework shows facility type ($\hat{\beta} = 6.38$) dominates income ($\hat{\beta} = 0.20$) by a factor of 30 — a finding invisible to aggregate accounts.

5.2 Pandey et al. (2018) - WHO Bulletin

Pandey, Ploubidis, Clarke, and Dandona analysed four National Sample Survey (NSS) consumer expenditure surveys and three utilisation surveys spanning 1993–2014 in the Bulletin of the World Health Organization [8]. They define CHE as OOP $\geq$ 10% of household expenditure - the same threshold we use — and report a rise from 12.4% (1993–94) to 24.9% by 2014. Rural households faced significantly higher CHE (aOR = 1.27, 95% CI: 1.20–1.35) and elderly-only households faced the highest risk (aOR = 3.26).

Convergence. Our study confirms both patterns. Rural households in our data show a catastrophic rate of 48.5% versus 45.5% for urban households. The age 60+ coefficient in our OLS model is $\hat{\beta} = 0.168$ ($p < 0.001$), consistent with higher OOP among the elderly.

Key divergence - Gender. Pandey finds female-headed households face significantly higher CHE (aOR = 1.32). We find no significant gender difference in log-OOP ($t = -0.254$, $p = 0.799$). Our individual-level model controls for facility type and illness severity; the female-head effect in Pandey may be confounded by differential access to care rather than higher OOP per episode.

What our study adds. Pandey uses descriptive and basic logistic analysis. We apply LASSO regularisation across 33 predictors - identifying facility type as the dominant predictor, a dimension Pandey's NSS data does not capture at the episode level.

5.3 Verma et al. (2021) - Equity in Health

Verma, Nath, and Tripathi use NSS health and consumption surveys from 2004, 2014, and 2018 to estimate CHE using the capacity-to-pay method [9]. They find CHE incidence fell from 13.4% (2014) to 9.1% (2018) and that health insurance reduced CHE only among the richest households.

Convergence. Both studies agree that income provides strong protection: Verma's richest quintile OR = 0.22; our Q4 OR = 0.494 versus Q1. Both find insurance effectiveness is concentrated among higher-income groups.

Divergence. Verma finds insurance reduces CHE; our overall logistic model finds insurance *increases* catastrophic odds (OR = 1.53, $p < 0.001$). This is not a contradiction. Our AA4 cross-tabulation shows insured households use private facilities less (28.1% vs. 39.3%; $\chi^2 = 407.8$, $p < 0.001$), ruling out moral hazard. The mechanism in our data is adverse selection: sicker individuals enrol in insurance and incur high OOP regardless, because reimbursement ceilings and outpatient exclusions leave most costs uncovered.

What our study adds. Verma cannot test the mechanism because NSS data do not identify facility type at the episode level. Our episode-level interaction model shows the private premium nearly doubles for inpatient versus outpatient care ($\hat{\beta}_3 = 1.17$, $p = 6.5 \times 10^{-132}$)

5.4 Garg et al. (2024) - BMC Medicine

Garg and colleagues conducted two annual household surveys (2021 and 2022) in Chhattisgarh, covering approximately 15,000 individuals each year, to evaluate AB-PMJAY's impact on financial protection [10]. They find that among AB-PMJAY enrolled individuals using private hospitals, the catastrophic rate at the 10% threshold was 78.1% (2021) and 70.9% (2022). Crucially, AB-PMJAY enrollment was not associated with reduced OOP or reduced catastrophic expenditure.

Convergence - Strongest in the Literature. Our Insured + Private cell catastrophic rate is 81.9% — virtually identical to Garg's 78.1%–70.9%. Both studies independently confirm that AB-PMJAY provides no meaningful financial protection at private facilities.

What our study adds. Garg is limited to one state. Our 29-state national evidence shows the insurance failure is geographically pervasive: the insurance effect on CHE ranges from -9.6 to $+39.5$ percentage points across states. We also formally test and rule out facility selection as the mechanism (AA4), and identify NCD patients as the most financially exposed subgroup - a finding Garg does not address.

5.4.1 Head-on State Comparison: Chhattisgarh Survey Validation

Our Chhattisgarh subsample ($n = 1,604$) provides a localized validation of the statewide longitudinal panel study conducted by Garg, Bebarta, and Tripathi (2024). Both analyses evaluate the AB-PMJAY program four years into its implementation, specifically following the 2020 upward revision of provider reimbursement rates.

Results and Synthesis. Our analysis identifies an 89.8% catastrophic rate specifically for insured households utilizing private facilities, a figure that aligns with the range of 70.9%-78.1% reported in the statewide study. This convergence confirms a critical failure in the scheme's design: private hospitals contracted under the scheme continue to overcharge patients, rendering purchasing ineffective at regulating provider behavior. While our data highlights a "catastrophic plateau" in the bottom half of the consumption distribution, the benchmark study establishes that enrollment is not associated with any significant reduction in overall out-of-pocket expenditure (OOPE) or catastrophic health expenditure (CHE).

Furthermore, both studies independently rule out moral hazard as a driver of high costs. Our data shows nearly identical private facility usage rates between insured (10.70%) and uninsured (10.79%) households, while the statewide survey utilizes propensity score matching (PSM) and instrumental variable (IV) methods to confirm that enrollment does not increase the utilization of inpatient care. Instead, the evidence points toward a persistent problem of "double-billing" and unauthorized copayments in the private sector.

What our study adds. While the benchmark study expands the evaluation to include quality metrics - finding that AB-PMJAY does not improve patient satisfaction or reduce the length of stay - our study provides deeper clinical and geographic context. We identify Non-Communicable Diseases (NCDs) as a primary driver of financial vulnerability, with insured NCD patients facing a 71.1% catastrophic rate, a subgroup vulnerability not explicitly modeled in the statewide survey. Moreover, by scaling this analysis across 29 states, we demonstrate that the "Chhattisgarh Model" is representative of a national "Insurance Paradox," where the depth of coverage ($\geq$ 75% from Both Studies) fails to match the aggressive pricing of private providers regardless of regional enrollment breadth.

Graph (A) in Appendix A, shows the head-on comparison.

Clarification on Metric Scaling. It is critical to distinguish between the statewide average catastrophic rate (∼39.8%) and the isolated "Insured + Private" cell (∼89.8%). The former is a blended metric reflecting Chhattisgarh's relatively robust public health infrastructure, which acts as a financial buffer for the majority of the population. The latter, however, isolates the specific intersection of adverse selection and private-sector overcharging. By focusing on this sub-segment, we reveal that for the sicker, private-facility-reliant population, insurance provides virtually no protection against aggressive pricing. This confirms that "breadth of enrollment" in the state has not yet translated into "depth of protection" for high-intensity care.

6 LIMITATIONS

Cross-Sectional Design. This dataset captures a single point in time for each household. No causal claims can be made. In particular, the OR = 1.53 finding for insurance may reflect adverse selection (sicker individuals enrolling) rather than insurance causing higher OOP. Establishing causation requires longitudinal data or a natural experiment.

Consumption Proxy. Total consumption is approximated by food plus education expenditure, excluding housing, transport, and utilities. This may underestimate economic capacity for urban households where non-food costs are substantial, potentially biasing consumption-related estimates.

Endogeneity of Facility Choice. Facility type is included as a predictor in RQ1, but facility choice and OOP are jointly determined. The coefficient on private facility use should be interpreted as a descriptive association, not a causal effect.

Missing UHC Data. The UHC index is unavailable for 3,800 respondents (7.6%), reducing the RQ5 sample to 46,200. If missingness is correlated with district characteristics, RQ5 findings may not generalise to the most under-resourced districts.

Recall Bias. Self-reported OOP is subject to recall bias. Smaller, frequent outpatient payments may be underreported relative to large hospitalisation costs, introducing a potential downward bias in outpatient OOP.

7 CONCLUSION

Central finding. OOP expenditure in India is determined overwhelmingly by *where* you seek care, not *who* you are. Private facility use adds $\beta = 6.38$ to log-OOP - an effect 30 times larger than the income gradient. Policies focused exclusively on expanding insurance coverage without addressing private sector pricing will not meaningfully reduce OOP for most households.

The Insurance Paradox. Insured households paradoxically face higher catastrophic odds ($OR = 1.53$). Facility selection is ruled out as the mechanism (AA4). Adverse selection - sicker individuals enrolling in schemes with low reimbursement ceilings and outpatient exclusions - is the most defensible explanation. The income quartile gradient ($OR = 0.494$ for Q4 vs. Q1) provides stronger protection than insurance, suggesting economic capacity is currently a more reliable buffer than formal coverage.

Threshold and Geography. Catastrophic risk is flat across Q1 (50.7%) and Q2 (50.8%) before declining at Q3 and Q4 - the bottom half of the consumption distribution faces identical exposure. State-level variation spans 74 percentage points (20.6% to 94.5%), and **district infrastructure quality independently reduces OOP by 4.3% per UHC index unit**.

Equity Failures. Chronic illness patients face higher OOP ($OR = 2.11$) and are less likely to be insured (24.3% vs. 27.7%). SC/ST households appear to spend less but seek less care - their lower OOP partly reflects foregone treatment rather than financial advantage.

Policy Recommendations:
Four targeted interventions follow from these findings:

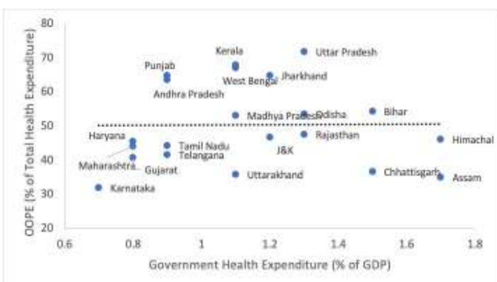
1. **Insurance reimbursement ceilings must be raised and outpatient coverage extended** - depth of coverage, not breadth of enrollment, determines financial protection.

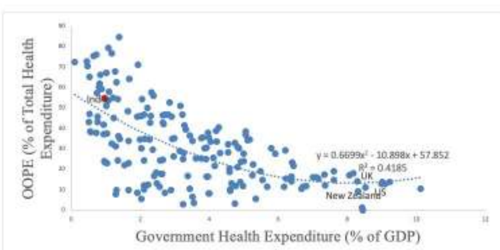
2. **Private sector price transparency and regulation** would address the dominant driver of OOP more directly than insurance expansion.

3. **Chronic illness patients need disease - specific financial protection mechanisms**, since they bear the highest risk and the lowest coverage.

4. **District infrastructure investment should be recognized as a financial protection tool**: better-resourced districts visibly produce measurably lower OOP for their residents **independent of household characteristics**.

Insurance Paradox - Breadth without Depth:





The macro-level graphs contrast rising public health spending with stagnant household-level financial protection. Despite government expenditure increasing to 48.0% of total health spending, the negligible correlation between insurance and OOP levels ($r = -0.005$) exposes a disconnect between national trends and individual reality. These findings suggest that OOP intensity is overwhelmingly determined by facility choice, indicating that current insurance instruments provide insufficient depth against private sector pricing and chronic illness shocks.

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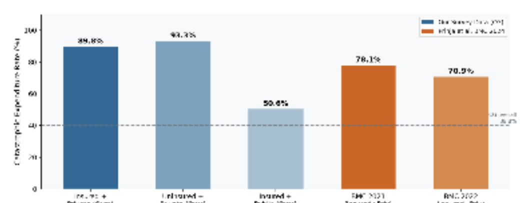
Appendix A - Head-On Comparison



The graph shows a comparison of the CHE Rate in both the studies, again showing that Private facility usage is a major factor despite having any form of insurance too.